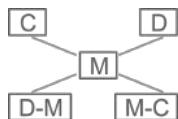


Mixture and Alligation

Mixture : when two or more than two substances mixed together in any ratio to produce a product is called mixture .

Mean price : the average of all the given prices of the known items is called Mean Price .

1. Alligation rule : It is used to find the ratio in which two or more substances at their respective prices should be mixed to produce a mixture at a given price.



where

C = Cost price of cheaper

D = Cost price of dearer

M = Mean price

(D - M) = Cheaper quantity

(M - C) = Dearer quantity

2. Quantity of pure liquid = $x\left(\frac{x-y}{x}\right)^n$

where, x = Quantity of original liquid, y = Quantity to be carried out and replaced, n = Number of times the operation is conducted

Important Cases :-

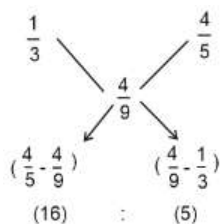
Case 1 :

Ratio of milk and water in container A, B and C are 1 : 2, 4 : 1 and 4 : 5 respectively. Find the ratio in which containers A and B are mixed to get 4 : 5 in container C ?

Soln : Fraction of milk in container A = $\frac{1}{3}$

Fraction of milk in container B = $\frac{4}{5}$

Fraction of milk in container C = $\frac{4}{9}$



So, the required ratio = 16 : 5

Case 2 :

Ratio of milk and water in container A and B are 3 : 1 and 4 : 1 respectively. Ratio in which container A and B are to be mixed is 2 : 1. Find the ratio of quantity of milk and water in container C.

Soln :

Fraction of milk in container A = $\frac{3}{4}$

Fraction of milk in container B = $\frac{4}{5}$

Ratio in which container A and B are to be mixed is 2 : 1. So, let the quantity in container A and B be 2 litre and 1 litre respectively.

So, the required ratio =

$$\left[\frac{3}{4} \times 2\right] + \left[\frac{4}{5} \times 1\right] : \left[\frac{1}{4} \times 2\right] + \left[\frac{1}{5} \times 1\right]$$

$$= \frac{3}{2} + \frac{4}{5} : \frac{1}{2} + \frac{1}{5} = \frac{23}{10} : \frac{7}{10} = 23 : 7$$

Case 3 : Two varieties costing ₹4/kg and ₹5/kg respectively are mixed in 3 : 2. Find the cost price per kg of the resultant mixture ?

Soln:

$$A_{wt} = \frac{A1W1 + A2W2}{W1 + W2} = \frac{4 \times 3 + 5 \times 2}{3 + 2}$$

$$= \frac{12 + 10}{5} = \frac{22}{5} = 4.4 \text{ kg}$$

Variety Questions

Q.1. A drum contains a blend of three chemicals: P, Q, and R, in the ratio of 3 : 5 : 2, 60 litres of this blend are removed, and then 12 litres of chemical P and 8 litres of chemical Q are poured into the drum. If the resultant quantity of chemical Q is 40 litres more than the resultant quantity of chemical P, what was the initial total quantity of the mixture in the drum (in litres) ?

SSC CGL 14/09/2025 (Shift 2)

(a) 300 (b) 320 (c) 280 (d) 380

Q.2. A milkman mixes water in pure milk and offers the mixture for sale at the original cost price of milk. If he mixes 0.25 liters of water in every 6.50 liters of milk and sells the entire mixture at the price of pure milk, what is his profit percentage?

Graduate Level 29/08/2025 (Shift - 1)

(a) 2.5% (b) 3.84% (c) 6.66% (d) 7.64%

Q.3. A 45-litre mixture of petrol and kerosene is in the ratio 5 : 4. Some of it is removed and replaced with pure petrol. The final ratio becomes 7 : 4. How much of the mixture was replaced ?

Graduate Level 29/07/2025 (Shift - 3)

(a) 10.6 litres (b) 9.78 litres
(c) 12.4 litres (d) 8.2 litres

Q.4 A solution of phenyl and water is 28 litres, with phenyl and water in ratio 4 : 3. 21 litres phenyl-water solution which has phenyl to water ratio as 2 : 1 is added to the 28 litres phenyl-water solution. Again 51 litres phenyl-water solution that has phenyl to water ratio as 9 : 8 is added to this. What is ratio of water to phenyl in the final mixture ?

(a) 57 : 43 (b) 59 : 41
(c) 41 : 59 (d) 43 : 57

Q.5. Two different quantities of the same solution having ingredients A and B are stored in two different containers. In the first container, there are 252 litre of A and 441 litre of B. In the second container, the total quantity of the solution was 1188 litre. How much of the solution in the second container was made up of ingredient B ?

(a) 765 litre (b) 756 litre
(c) 752 litre (d) 760 litre

Q.6. A bucket contains a mixture of milk and water in the proportion of 7 : 5. If 9 litres of milk is replaced by water, then the ratio becomes 7 : 9. How much milk was there in the bucket initially ?

(a) 36 litres (b) 30 litres
(c) 25 litres (d) 35 litres

Q.7. A drink is made by mixing water and juice in the ratio 9 : 7. If x litres of water and $2x$ litres of juice are mixed in 160 litres of drink, then the new ratio becomes 13 : 15. The quantity of drink (in litres) is:

(a) 240 (b) 280 (c) 300 (d) 260

Q.8. 80 litres of a mixture of spirit and water in the ratio 7 : 9 is present in a container A. 20 litres of the mixture is transferred to another container B. Then the container A is filled with 20 litres of water. Then 32 litres of the mixture is again transferred to container B. The ratio of water to spirit of the mixture in container B is equal to:

(a) 41 : 87 (b) 87 : 41
(c) 131 : 77 (d) 77 : 131

Q.9. At a certain time in a park, the number of heads and the number of legs of monkeys and human visitors were counted, and it was found that there were 54 heads and 148 legs. Find the number of monkeys in the park.

(a) 20 (b) 16 (c) 18 (d) 14

Q.10. A milk vendor has 2 vessels of milk and water. The first contains 80% milk and the rest is water. The second contains 50% milk. How much mixture (in L) should he mix from each of the vessels so as to get 9 litres of mixture such that the ratio of water to milk is 1 : 2 ? (Write your answer in the order of vessel 1 and vessel 2)

(a) 6, 3 (b) 4, 5 (c) 3, 6 (d) 5, 4

Q.11. A blended tea in a barrel contains Assam, Darjeeling and Nilgiri Hills in the ratio 2 : 3 : 1. Another barrel contains Darjeeling, Nilgiri Hills and Kangra Valley in the ratio 5 : 4 : 3. If equal weights of both blended tea mix together to make a



new brand, then the weight of Kangra Valley per kilogram in the new brand is:

- (a) 500 gm (b) 100 gm
(c) 250 gm (d) 125 gm

Q.12. From a cask of milk containing 64 liters, 8 liters are drawn out and the cask is filled up with water. If the same process is repeated a second, then a third time, what will be the proportion of milk to water in the resulting mixture ?

- (a) 343 : 179 (b) 333 : 169
(c) 443 : 179 (d) 343 : 169

Q.13. A jar contains 35% syrup in the mixture of syrup and water. A part of this mixture is replaced by another mixture containing 20% syrup. Now, in the new mixture, the percentage of syrup is found to be 24%. What is the ratio in which these mixtures were mixed to obtain the new mixture ?

- (a) 3 : 14 (b) 4 : 11 (c) 5 : 13 (d) 7 : 10

Q.14. In a solution of water and milk, the quantity of milk is double the quantity of water. To pass the quality standards, the solution must contain 90% milk in it. If the volume of the solution is v , then how much pure milk in litres is to be added to the solution to pass the quality standard.

- (a) $\frac{4}{3}v$ (b) $\frac{3}{4}v$ (c) $\frac{7}{4}v$ (d) $\frac{7}{3}v$

Q.15. Alloy X contains 70% copper and 30% zinc. Alloy Y contains 40% copper, 25% zinc and 35% aluminum. Alloy X and Y are mixed in the ratio of 1 : 3. What is the ratio of copper and zinc in the newly formed alloy ?

- (a) 19 : 33 (b) 11 : 32
(c) 38 : 21 (d) 19 : 21

Q.16. A jar contains a mixture of milk and water, having 50% water. A part of this mixture is replaced by another mixture having 29% water. The percentage of water in the jar after replacement is 36%. What fraction of the original mixture was replaced ?

- (a) $\frac{3}{2}$ (b) $\frac{4}{5}$ (c) $\frac{1}{3}$ (d) $\frac{2}{3}$

Q.17. Two vessels of equal capacity, and both completely filled, contain mixtures of milk and water in the ratio 5 : 1 and 6 : 5, respectively. Both the mixtures are now mixed thoroughly. What is the ratio of milk to water in the new mixture so obtained ?

- (a) 105 : 67 (b) 57 : 41
(c) 91 : 41 (d) 28 : 43

Q.18. Babulal runs a juice corner outside a park and sells giloy juice (giloy + water) in the morning. Initially he had 19 litres of

juice, which had giloy and water in the ratio 2 : 9. He sold $2\frac{1}{2}$ litres of juice.

Later, in order to dilute it, he added some water and the ratio of giloy and water became 1 : 5. How much water was added ?

- (a) $1\frac{1}{2}$ litres (b) $\frac{1}{2}$ litre
(c) 2 litres (d) 1 litre

Q.19. Alloy A contains metal x and y only in the ratio 5 : 2 and alloy B contains these metals in the ratio 3 : 4. Alloy C is prepared by mixing A and B in the ratio 4 : 5. The percentage of x in alloy C is:

- (a) $44\frac{4}{9}$ (b) 56 (c) 45 (d) $55\frac{5}{9}$

Q.20. Two bottles of same capacity are 35% and $33\frac{1}{3}$ % full of orange juice, respectively. They are filled up completely with apple juice and then the contents of both bottles are emptied into another vessel. The percentage of apple juice in the mixture is:

- (a) $34\frac{1}{6}$ (b) $64\frac{1}{3}$ (c) $60\frac{2}{3}$ (d) $65\frac{5}{6}$

Q.21. A vessel contains a 32 litre solution of acid and water in which the ratio of acid and water is 5 : 3. If 12 litres of the solution are taken out and $7\frac{1}{2}$ litres of water are added to it, then what is the ratio of acid and water in the resulting solution ?

- (a) 4 : 7 (b) 5 : 6 (c) 4 : 9 (d) 8 : 11

Q.22. The ratios of copper to Zinc in alloys A and B are 3 : 4 and 5 : 9 respectively. A and B are taken in the ratio 2 : 3 and melted to form a new alloy C. What is the ratio of copper to Zinc in C ?

- (a) 8 : 13 (b) 3 : 5 (c) 9 : 11 (d) 27 : 43

Q.23. A milkman uses three containers for selling milk, their capacities being 40L, 30L and 20L respectively. He fills respectively 87.5%, 80% and 90% of the containers with a mix of milk and water in the ratios, 3 : 2, 5 : 1 and 7 : 2 respectively. What is the ratio of total quantity of milk to that of water carried by him ?

- (a) 7 : 2 (b) 31 : 12 (c) 35 : 9 (d) 5 : 2

Q.24. Fresh fruit contains 68% water and dry fruit contains 20% water. How much dry fruit can be obtained from 100 kg of fresh fruit

- (a) 80 (b) 60 (c) 40 (d) 20

Examination wise Questions

SSC CGL 2025 Tier - 1

Q.25. A 80-liter mixture contains Milk and water in the ratio 5 : 3. How much water (in liters) must be added to this mixture to change the ratio of Milk to water to 2 : 3 ?

SSC CGL 12/09/2025 (Shift 1)

- (a) 40 litres (b) 45 litres
(c) 35 litres (d) 30 litres

Q.26. Two Lentils priced at ₹90 per kg and ₹220 per kg are blended and sold at ₹250 per kg, achieving a profit margin of 25%. What is the ratio of the two lentils in the mixture ?

SSC CGL 12/09/2025 (Shift 1)

- (a) 2 : 11 (b) 11 : 2 (c) 9 : 2 (d) 11 : 1

Q.27. A 200-liter solution contains milk and water in the ratio 3 : 2. Some of this solution is removed and replaced with pure milk. If the final ratio of milk and water becomes 7 : 3, how many liters of solution were replaced ?

SSC CGL 12/09/2025 (Shift 2)

- (a) 35 litres (b) 25 litres
(c) 50 litres (d) 20 litres

Q.28. Approximately how many kg of pulses costing ₹45 per kg should be mixed with 24 kg of pulses costing ₹32 per kg so as to make a profit of 15% by selling the mixture at ₹43.70 per kg ?

SSC CGL 12/09/2025 (Shift 3)

- (a) 17.3 kg (b) 22.4 kg
(c) 20.5 kg (d) 15.2 kg

Q.29. From a 64-liter container of pure milk, 8 liters are removed and replaced by water. This operation is performed a total of three times. What is the final quantity of milk remaining in the container ?

SSC CGL 13/09/2025 (Shift 1)

- (a) 44.44 liters (b) 55.45 liters
(c) 40.42 liters (d) 42.88 liters

Q.30. A juice vendor mixes water with pure mango juice. If he sells the mixture at the cost price of pure mango juice and makes a profit of 20%, what is the ratio of water to pure mango juice in the mixture, respectively ?

SSC CGL 13/09/2025 (Shift 1)

- (a) 3 : 2 (b) 1 : 5 (c) 2 : 3 (d) 5 : 1

Q.31. A 70 litre mixture contains milk and water in the ratio 4 : 3. If 7 litres of this mixture is removed and 10 litres of pure milk is added in its place, what will be the



new ratio of milk and water in the mixture?

SSC CGL 13/09/2025 (Shift 2)

- (a) 23 : 18 (b) 46 : 27
(c) 46 : 21 (d) 43 : 27

Q.32. A 30-litres solution of alcohol and water has 20% water. How many litres of alcohol must be added to the solution to make the water content 15% ?
SSC CGL 14/09/2025 (Shift 1)

- (a) 10 litres (b) 20 litres
(c) 30 litres (d) 40 litres

Q.33. Two containers of the same volume are 30% and 50% full of milk, respectively. They are then filled completely with water. If the contents of both containers are mixed in a larger vessel, what is the percentage of water in the final mixture ?
SSC CGL 14/09/2025 (Shift 3)

- (a) 45% (b) 50% (c) 55% (d) 60%

Q.34. A solution consists of milk and water in the ratio 3 : 5. If 8 litres of water is added, the new ratio becomes 3 : 7. What is the original quantity of milk in the solution ?
SSC CGL 14/09/2025 (Shift 3)

- (a) 12 litres (b) 15 litres
(c) 20 litres (d) 30 litres

Q.35. A tank has a mixture of solutions A, B, and C in the respective ratio of 5 : 3 : 2. 10 litres of this mixture is drained out, and subsequently, 5 litres of solution A and 3 litres of solution C are added to the tank. If the resultant quantity of solution A is 25 litres more than the resultant quantity of solution B, what was the initial quantity of mixture in the tank (in litres) ?
SSC CGL 14/09/2025 (Shift 3)

- (a) 140 (b) 240 (c) 120 (d) 110

Q.36. A container holds a mixture of three liquids P, Q, and R in ratio 4 : 6 : 10. If 6 liters of liquid P, 12 liters of liquid Q, and a certain amount of liquid R are added to the container, the new ratio of liquids P, Q, and R becomes 6 : 10 : 16. Find the quantity (in liters) of liquid R added.
SSC CGL 15/09/2025 (Shift 1)

- (a) 12 liters (b) 18 liters
(c) 16 liters (d) 10 liters

Q.37. A cistern contains 200 liters of pure syrup. 10 liters of syrup are drawn out and replaced by water. Then, 10 liters of the mixture are drawn out and replaced by water again. What is the ratio of water to syrup in the final mixture ?
SSC CGL 15/09/2025 (Shift 1)

- (a) 19 : 1 (b) 19 : 14
(c) 19 : 81 (d) 18 : 19

Q.38. A container holds a mixture of Mango, Apple, and Pomegranate juices in the ratio of 5 : 7 : 4 respectively. 24 litres of this mixture is taken out, and then 10 litres of Mango juice and 6 litres of Pomegranate juice are added to the vessel. If the resultant quantity of apple juice is 14 litres more than the resultant quantity of Mango juice, what was the initial quantity of mixture in the vessel (in litres) ?
SSC CGL 15/09/2025 (Shift 3)

- (a) 216 (b) 210 (c) 214 (d) 235

Q.39. A tank holds $7\frac{3}{4}$ liters of water. If 3.375 liters are drained twice, how much water remains ?
SSC CGL 16/09/2025 (Shift 1)

- (a) 1 (b) $2\frac{1}{4}$ (c) $3\frac{3}{8}$ (d) $\frac{1}{2}$

Q.40. Two containers, A and B, contain mixtures of alcohol and water. Container A has alcohol and water in the ratio 2 : 1, while Container B has them in the ratio 4 : 3. If 9 liters are drawn from Container A and 14 liters from Container B, and the contents are mixed in a third container, what is the ratio of alcohol to water in the new mixture ?
SSC CGL 16/09/2025 (Shift 1)

- (a) 3 : 11 (b) 14 : 9 (c) 9 : 14 (d) 11 : 3

Q.41. Two distinct varieties of wheat, valued at ₹80 per kilogram and ₹120 per kilogram respectively, are combined in an unknown ratio. The resulting mixture is sold at ₹130 per kilogram, yielding a profit of 25% on the aggregate cost price. Determine the ratio in which the two wheat varieties were blended.
SSC CGL 16/09/2025 (Shift 2)

- (a) 1 : 3 (b) 3 : 4 (c) 2 : 3 (d) 4 : 1

Q.42. A 120 - litre mixture contains milk and water in the ratio 5 : 3. If 'x' litres of milk and 'x' litres of water are added to the mixture, the new ratio of milk to water becomes 3 : 2. What is the value of 'x' ?
SSC CGL 16/09/2025 (Shift 3)

- (a) 12.33 litres (b) 11.54 litres
(c) 15 litres (d) 30 litres

Q.43. A barrel contains a mixture of alcohol and water in the ratio 5 : 3 respectively. If 24 litres of water are added to it, the ratio of alcohol to water becomes 5 : 9. Find the initial total quantity of the mixture in the barrel.
SSC CGL 17/09/2025 (Shift 2)

- (a) 32 litres (b) 40 litres

- (c) 52 litres (d) 60 litres

Q.44. Two containers of equal capacity filled with juice and water. The first contains juice and water in the ratio 2 : 3, and the second in the ratio 3 : 2. The mixtures are then combined. What is the ratio of juice to water in the resulting mixture ?
SSC CGL 18/09/2025 (Shift 3)

- (a) 1 : 1 (b) 3 : 4 (c) 4 : 3 (d) 5 : 4

Q.45. A Barthan has 90 litres of a solution with sugar and soda in the ratio 7 : 2. If 18 litres of the solution are replaced with soda, what is the new ratio ?
SSC CGL 20/09/2025 (Shift 1)

- (a) 20 : 17 (b) 28 : 17
(c) 31 : 15 (d) 13 : 23

Q.46. In a 60-liter mixture of apple juice and water, the amount of water is 20 liters. How many liters of water should be added to make the ratio of juice to water 2 : 3 ?
SSC CGL 20/09/2025 (Shift 3)

- (a) 50 L (b) 30 L (c) 40 L (d) 80 L

Q.47. A vessel contains milk and water in the ratio 6 : 5. If 10 liters of milk are removed and 9 liters of water are added, the new ratio becomes 2 : 3. What was the original quantity of the mixture in the vessel ?
SSC CGL 20/09/2025 (Shift 3)

- (a) 67 (b) 66 (c) 68 (d) 78

Q.48. A mixture of 200 litres contains acid and water in the ratio 3:5. If 40 litres of the mixture is replaced with water, what will be the percentage of acid in the final mixture ?
SSC CGL 21/09/2025 (Shift 2)

- (a) 29% (b) 28% (c) 32% (d) 30%

Q.49. A 32-litre mixture of milk and water contains milk and water in the ratio 3 : 5. How much water must be added to make the ratio 3 : 6 ?
SSC CGL 22/09/2025 (Shift 1)

- (a) 3 L (b) 4 L (c) 5 L (d) 7.5 L

Q.50. An alloy contains gold and silver in the ratio 5 : 3. A second alloy contains gold and silver in the ratio 3 : 1. What quantity of the first alloy must be added to 40 kg of the second alloy to form a new alloy containing gold and silver in the ratio 4 : 2 ?
SSC CGL 22/09/2025 (Shift 2)

- (a) 20 kg (b) 60 kg (c) 80 kg (d) 40 kg

Q.51. A merchant wants to mix two types of tea priced at Rs. 400/kg and Rs.



340/kg. In what ratio should they be mixed to sell the mixture at Rs. 370/kg without profit or loss ?

SSC CGL 22/09/2025 (Shift 3)

- (a) 1 : 1 (b) 2 : 1 (c) 3 : 2 (d) 1 : 2

Q.52. A solution of 150 litres has a 40% salt concentration. How many litres of water must be added to reduce the concentration to 20%, and then how much pure salt must be added to bring the concentration back to 50% ?

SSC CGL 23/09/2025 (Shift 2)

- (a) Add 150 L of water, then 180 L of salt
(b) Add 90 L of water, then 170 L of salt
(c) Add 170 L of water, then 30 L of salt
(d) Add 150 L of water, then 10 L of salt

Q.53. Two numbers are in the ratio 8 : 9. If 8 is added to each, the new ratio becomes 17 : 19. What are the original numbers ?

SSC CGL 24/09/2025 (Shift 1)

- (a) 143 and 179 (b) 128 and 144
(c) 115 and 137 (d) 136 and 152

Q.54. A container has a 80-litre mixture of milk and water in the ratio 3 : 1. 12 litres is removed and replaced with water. Then 12 litres is removed again and replaced with milk. What is the final percentage of water ?

SSC CGL 24/09/2025 (Shift 1)

- (a) 21.25% (b) 30.81%
(c) 25.25% (d) 20.25%

Q.55. A 40-litre mixture of milk and water contains milk and water in the ratio 3 : 1. Some quantity of water is added, changing the ratio to 3 : 5. What is the quantity of water added ?

SSC CGL 24/09/2025 (Shift 3)

- (a) 20 L (b) 40 L (c) 35 L (d) 25 L

Q.56. A 100-litre container of pure milk undergoes a 10-litre removal and 10-litre water replacement process, repeated 3 times. Approximately how much original milk remains?

SSC CGL 25/09/2025 (Shift 2)

- (a) $15\frac{78}{255}$ litres (b) $17\frac{29}{100}$ litres
(c) $8\frac{29}{100}$ litres (d) $72\frac{9}{10}$ litres

Q.57. A container has 100 litres of a mixture of acid and water in the ratio 3 : 2. 10% of the mixture is taken out and replaced with water. This operation is repeated once more. What is the final percentage of acid in the mixture ?

SSC CGL 26/09/2025 (Shift 1)

- (a) 54.8% (b) 41.2% (c) 43.7% (d) 48.6%

Q.58. A bag contains red, blue, and green marbles in the ratio 4 : 5 : 6. If 6 red

marbles are removed and 3 green marbles are added, the new ratio becomes 2 : 5 : 7. How many red marbles were originally in the bag ?

SSC CGL 26/09/2025 (Shift 2)

- (a) 12 (b) 14 (c) 16 (d) 20

Q.59. A tank contains 150 litres of liquid, 30% of which is alcohol. If 20 litres of this mixture is replaced by another liquid containing 40 % alcohol, what is the percentage of alcohol in the resulting solution ?

SSC CGL 26/09/2025 (Shift 2)

- (a) 31.33% (b) 35% (c) 36.33% (d) 31%

Q.60. A 80-litres mixture of acid and water contains acid and water in the ratio 4 : 1. 10 litres of this mixture is removed and replaced with another solution of acid and water in the ratio 1 : 4. What is the final ratio of acid to water in the resulting mixture ?

SSC CGL 26/09/2025 (Shift 3)

- (a) 29 : 11 (b) 11 : 5 (c) 11 : 29 (d) 17 : 13

Q.61. A 136 - litre container of pure milk undergoes a 17-litre removal and 17-litre water replacement process, repeated 2 times. Approximately how much original milk remains?

SSC CGL 14/10/2025 (Shift 1)

- (a) $94\frac{1}{8}$ litres (b) $104\frac{2}{8}$ litres
(c) $94\frac{2}{8}$ litres (d) $104\frac{1}{8}$ litres

Q.62. From a 54-liter container of pure spirit, 9 liters are removed and replaced by water. This operation is performed a total of three times. What is the final quantity of spirit remaining in the container ?

SSC CGL 14/10/2025 (Shift 3)

- (a) 31.34 liters (b) 36.45 liters
(c) 31.25 liters (d) 42.88 liters

Q.63. A cistern contains 100 liters of pure milk. 20 liters of milk are drawn out and replaced by water. Then, 20 liters of the mixture are drawn out and replaced by water again. What is the ratio of water to milk in the final mixture ?

SSC CGL 14/10/2025 (Shift 3)

- (a) 18 : 1 (b) 9 : 16 (c) 16 : 9 (d) 1 : 18

SSC Selection Post (XIII)

Q.64. Malika bought two varieties of pulses, costing Rs. 75 per kg and Rs. 125 per kg each, and mixed them in some ratio. Then she sold the mixture at Rs. 121 per kg. Making a profit of 10 percent. What was the ratio of the mixture?

Matriculation Level 30/07/2025(Shift - 1)

- (a) 3 : 7 (b) 1 : 5 (c) 7 : 3 (d) 4 : 11

Q.65. Two varieties of rice are mixed. The first variety costs ₹60 per kg and the second variety costs ₹90 per kg. In what ratio should these be mixed so that by selling the mixture at ₹84 per kg, a profit of 20% is gained?

Graduate Level 24/07/2025 (Shift - 3)

- (a) 1 : 2 (b) 2 : 1 (c) 3 : 2 (d) 4 : 3

Q.66. A merchant mixes 80 kg of coffee at ₹100/kg with a dearer variety at ₹140/kg. He sells the blend at ₹138/kg and gains 15% profit. How many kg of the dearer variety did he add?

Graduate Level 25/07/2025 (Shift - 3)

- (a) 80 kg (b) 88 kg (c) 85 kg (d) 87 kg

Q.67. A cistern contains 120 liters of pure milk. 30 liters of milk are drawn out and replaced by water. Then, 30 liters of the mixture are drawn out and replaced by water again. What is the ratio of water to milk in the final mixture ?

Graduate Level 26/07/2025 (Shift - 3)

- (a) 4 : 5 (b) 5 : 4 (c) 7 : 9 (d) 9 : 7

Q.68. A solution of 150 liters contains 30% acid. How much pure acid (in liters) must be added to make the solution 50% acid?

Graduate Level 28/07/2025 (Shift - 3)

- (a) 60 liters (b) 50 liters
(c) 36 liters (d) 24 liters

Q.69. A shopkeeper mixes 45 kg of rice at ₹65/kg with another grade at ₹95/kg. He sells the mixture at ₹80/kg, making a profit of $14\frac{1}{7}$ %. How many kg of the dearer rice did he use?

Graduate Level 02/08/2025 (Shift - 3)

- (a) 45 kg (b) 9 kg (c) 25 kg (d) 5 kg

Q.70. A mixture contains juice and water in the ratio 4 : 3. If 21 litres of water are added to this mixture, the ratio of juice to water becomes 5 : 4. What was the initial quantity of the mixture (in litres)?

Graduate Level 31/07/2025 (Shift - 3)

- (a) 165 litres (b) 176 litres
(c) 147 litres (d) 190 litres

Q.71. A 60-liter mixture contains acid and water in the ratio 7 : 3. How much water (in liters) must be added to this mixture to change the ratio of acid to water to 4 : 5?

Graduate Level 29/08/2025 (Shift - 1)

- (a) 25.5 litres (b) 34.5 litres
(c) 45 litres (d) 20 litres



eduquity-based pattern (ebp) Questions

Q.72. If two kinds of resins costing ₹160 per kg and ₹210 per kg are mixed in the ratio of 2 : 3, then find the cost (in ₹) of the mixture per kg.

(a) 185 (b) 190 (c) 200 (d) 180

Q.73. A chef has two cans of acids and water mixture. He combines the contents of both cans in a beaker. The resulting mixture has half water and half acid. What is the quantity in which he mixed the contents of can 1 and can 2, given that can 2 has an acid-to-water ratio of 2 : 3 and can 1 has an acid-to-water ratio of 5 : 3 ?

(a) 2 : 3 (b) 4 : 5 (c) 4 : 7 (d) 3 : 7

Q.74. A vessel contains 80 litres of a mixture of alcohol and water in the ratio of 3 : 7. If 40 litres of the mixture is taken out from the vessel and replaced with 40 litres of water, then the ratio of alcohol and water becomes:

(a) 3 : 17 (b) 1 : 11 (c) 3 : 23 (d) 2 : 15

Q.75. A can contains a mixture of two liquids, grape juice and pineapple juice in the ratio 13 : 11. When 8 litres of the mixture is drawn off, and the can is filled with pineapple juice, the ratio becomes 13 : 14. How many litres of pineapple juice was initially contained in the can ?

(a) 14 litres (b) 28 litres
(c) 11 litres (d) 33 litres

Q.76. A shopkeeper of a subsidised canteen mixed three varieties of rice, X, Y and Z, in the ratio of 4 : 1 : 3 to make a packet of 4 kg of rice. The cost prices per kg of X, Y and Z varieties of rice are ₹2, ₹4 and ₹6, respectively. What is the cost price (in ₹) of the packet of rice ?

(a) 10 (b) 15 (c) 30 (d) 18

Q.77. Rice worth ₹84 per kg and ₹96 per kg is mixed with a third variety of rice in the ratio 1 : 1 : 2. If the mixture is worth ₹99 per kg, the price of the third variety per kg is:

(a) ₹108 (b) ₹115 (c) ₹112 (d) ₹110

Q.78. The concentrations of three acids, A, B, and C, are specified as 20%, 30%, and 40%, respectively. They are blended in a ratio of 3 : 5 : a, yielding a solution with a concentration of 30%. What is the value of 'a' ?

(a) 3 (b) 1 (c) 2 (d) 4

Q.79. A merchant mixes two varieties of sugar of 5 kg and 3 kg, which costs him ₹4 and ₹6 per kg, respectively. What is

the cost price (in ₹) of the mixture per kg
(a) 4.35 (b) 4.45 (c) 4.75 (d) 4.21

Q.80. A shopkeeper mixes two types of rice costing ₹250/5 kg and ₹280/5 kg in the ratio 7 : 8. Find the net cost of the resultant mixture (per kg).

(a) ₹32.50 (b) ₹53.20
(c) ₹52.30 (d) ₹25.30

Q.81. 2 varieties of tea powders worth ₹250/kg and ₹300/kg are mixed with a third variety of tea powder in the ratio 1 : 1 : 2. If the mixture is worth ₹350/kg, find the price of the third variety tea powder.

(a) ₹450/kg (b) ₹400/kg
(c) ₹425/kg (d) ₹500/kg

Q.82. 15 kg of ₹10 per kg wheat is mixed with 5 kg of another type of wheat to get a mixture costing ₹30 per kg. Find the price (per kg) of the costlier wheat.

(a) ₹70 (b) ₹80 (c) ₹60 (d) ₹90

Q.83. A mango juice is made by mixing water and mango concentrate in the ratio 9 : 7. If x litres of water and 3x litres of mango concentrate is mixed in 160 litres of mango juice, then the new ratio becomes 13 : 14. What is the quantity of the new mango juice (in litres)?

(a) 197 (b) 212 (c) 206 (d) 216

Q.84. In what ratio, sugar costing ₹78 per kg be mixed with sugar costing ₹46 per kg such that by selling the mixture at ₹75 per kg there is a gain of 20% ?

(a) 33 : 37 (b) 31 : 29
(c) 37 : 31 (d) 33 : 31

Q.85. A man buys 10 kg of wheat at a rate of ₹26/kg. The wheat is mixed with 6 kg of other good quality of wheat to get a mixture at a price of ₹35/kg. The price of good quality wheat per kg (in ₹) is:

(a) 38 (b) 42 (c) 50 (d) 46

Q.86. Half litre of a solution contains 15% of alcohol. To change the alcohol concentration to 50%, find the quantity of alcohol to be mixed.

(a) 250 ml (b) 350 ml
(c) 175 ml (d) 400 ml

Q.87. How much water should be added to 60 liters of milk at $1\frac{1}{2}$ liters a rupee

so as to have a mixture worth $1\frac{7}{8}$ liters a rupee ?

(a) 15 liters (b) 20 liters
(c) 27 liters (d) 12 liters

Q.88. Ramu mixes 2 litres of water at 100° C and 18 litres of water at 32° C. What temperature will the water have

after mixing ?

(a) 40°C (b) 20°C (c) 38.8°C (d) 66°C

Answer Key :-

1.(c)	2.(b)	3.(d)	4.(a)
5.(b)	6.(a)	7.(b)	8.(c)
9.(a)	10.(d)	11.(d)	12.(d)
13.(b)	14.(d)	15.(c)	16.(d)
17.(c)	18.(a)	19.(d)	20.(d)
21.(b)	22.(d)	23.(d)	24.(c)
25.(b)	26.(a)	27.(c)	28.(c)
29.(d)	30.(b)	31.(b)	32.(a)
33.(d)	34.(a)	35.(d)	36.(b)
37.(c)	38.(a)	39.(a)	40.(b)
41.(c)	42.(c)	43.(a)	44.(a)
45.(b)	46.(c)	47.(b)	48.(d)
49.(b)	50.(c)	51.(a)	52.(a)
53.(b)	54.(b)	55.(b)	56.(d)
57.(d)	58.(a)	59.(a)	60.(a)
61.(d)	62.(c)	63.(b)	64.(a)
65.(b)	66.(a)	67.(c)	68.(a)
69.(b)	70.(c)	71.(b)	72.(b)
73.(b)	74.(a)	75.(d)	76.(b)
77.(a)	78.(a)	79.(c)	80.(b)
81.(c)	82.(d)	83.(d)	84.(d)
85.(c)	86.(b)	87.(a)	88.(c)

Solutions :-

Sol.1.(c) Let the quantity of P, Q and R be 3x, 5x and 2x respectively,
According to question,

$$\Rightarrow P : Q : R = (3x - 3 \times \frac{60}{10}) + 12 : (5x - 5 \times$$

$$\frac{60}{10}) + 8 : (2x - 2 \times \frac{60}{10})$$

$$\Rightarrow P : Q : R = (3x - 6) : (5x - 22) : (2x - 12)$$

Now,

$$\Rightarrow (5x - 22) - (3x - 6) = 40$$

$$\Rightarrow 2x = 56 \Rightarrow x = 28$$

So, the initial quantity of the mixture
= $28 \times (3 + 5 + 2) = 280$ Litres

Sol.2.(b)

Let Cost price of milk = ₹1 per litre

Total cost price = ₹6.50

Now he mix 0.25 L water with 6.50 L ,

Total mixture = $6.50 + 0.25 = 6.75$ L

Now total selling price = ₹6.75

$$\text{Profit percentage} = \frac{6.75 - 6.50}{6.50} \times 100$$

$$= 3.84\%$$

Sol.3.(d)



	P	:	K	
Ratio	→ 5	:	4	9 unit = 45L
Initial quantity	→ 25L	:	20L	

$$\text{Final quantity} \rightarrow 25L - \frac{5x}{9} + x : 20L - \frac{4x}{9}$$

A/Q,

$$\Rightarrow \frac{25 - \frac{5x}{9} + x}{20 - \frac{4x}{9}} = \frac{7}{4}$$

$$\Rightarrow 4(25 + \frac{4x}{9}) = 7(20 - \frac{4x}{9})$$

$$\Rightarrow 100 + \frac{16x}{9} = 140 - \frac{28x}{9}$$

$$\Rightarrow \frac{44x}{9} = 40 \Rightarrow x = 8.18 \text{ L} \approx 8.2 \text{ L}$$

Sol.4.(a) According to question,

Ratio of solution → phenyl : water

28 litres → 16 : 12

21 litres → 14 : 7

51 litres → 27 : 24

Final mixture → 57 : 43

Hence, Required ratio = 57 : 43

Sol.5.(b) In 1st container -

Ratio of ingredients A and B = 252 : 441

= 12 : 21

In 2nd container -

$$\text{Quantity of ingredient B} = \frac{21}{33} \times 1188$$

= 756 litre

Sol.6.(a) Let the fraction reduced by 'f'

$$\frac{7}{12} \times f = \frac{7}{16} \Rightarrow f = \frac{12}{16} = \frac{3}{4}$$

$$\text{Quantity of milk replaced by water} = \frac{1}{4}$$

According to question,

$$\frac{1}{4} = 9 \text{ litres}$$

So, the initial quantity of milk

= 4 × 9 = 36 litres

Sol.7.(b)

Mixture → Water : Juice

160 liter → 90 lit. : 70 lit.

↓ + x lit. : ↓ + 2x lit.

Final → 13 : 15

$$\frac{90 + x}{70 + 2x} = \frac{13}{15}$$

$$1350 + 15x = 910 + 26x$$

$$11x = 440 \Rightarrow x = 40$$

$$\text{Water + juice added} = x + 2x = 40 + 80 = 120 \text{ litres}$$

$$\text{Total quantity of drink} = 160 + 120$$

$$= 280 \text{ litres}$$

Sol.8.(c) when 20 litres of the mixture is transferred to another container B. then,

$$\text{Quantity of spirit in container B} = \frac{7}{16}$$

After replacement with 20 liter water, Quantity of spirit in

$$\text{container A} = \frac{7}{16} \times \frac{3}{4} = \frac{21}{64}$$

Again 32 litres of this mixture is transferred to container B.

Then,

$$\text{Quantity of spirit in container B} = \frac{21}{64}$$

So, the ratio of the quantity of two types of mixture in container B = 20 : 32 = 5 : 8

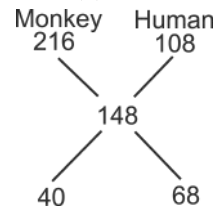
Now, using weighted average, we have ;

Quantity of spirit

$$= \frac{5 \times \frac{7}{16} + 8 \times \frac{21}{64}}{5 + 8} = \frac{308}{64 \times 13} = \frac{77}{208}$$

The ratio of water and spirit in container

$$B = 208 - 77 : 77 = 131 : 77$$

Sol.9.(a)

So, the ratio of the no of monkey and

human = 40 : 68 = 10 : 17

$$10 + 17 = 27 \text{ unit} = 54$$

$$10 \text{ unit} = \frac{54}{27} \times 10 = 20$$

Sol.10.(d) According to the question,

Ratio - milk : water

$$1^{\text{st}} \text{ vessel} - 4 : 1 = 5 \Rightarrow 2 \times 3 = 6$$

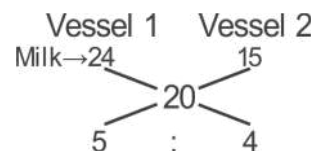
$$2^{\text{nd}} \text{ vessel} - 1 : 1 = 2 \Rightarrow 5 \times 3 = 15$$

$$3^{\text{rd}} \text{ vessel} - 2 : 1 = 3 \Rightarrow 2 \times 5 = 10$$

$$1^{\text{st}} \text{ vessel Milk} = 2 \times 3 \times 4 = 24$$

$$2^{\text{nd}} \text{ vessel Milk} = 5 \times 3 \times 1 = 15$$

$$3^{\text{rd}} \text{ vessel Milk} = 2 \times 5 \times 2 = 20$$



Required quantity = 5, 4

Sol.11.(d)

Ratio → Assam : Darjeeling : Nilgiri hills

$$\text{Tea} \rightarrow 2 : 3 : 1$$

$$\text{or } 4 : 6 : 2 = 12 \text{ unit}$$

Ratio → Darjeeling : Nilgiri : kangra

$$\text{Tea} \rightarrow 5 : 4 : 3 = 12 \text{ unit}$$

Ratio after blended tea are mixed :-

$$A : D : N : K = 4 : (6 + 5) : (2 + 4) : 3$$

$$= 4 : 11 : 6 : 3 = 24 \text{ units}$$

Weight of kangra valley tea per kg

$$= \frac{3}{24} \times 1000 = 125 \text{ gm}$$

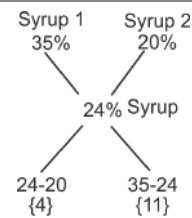
$$\text{Sol.12.(d)} \quad \frac{\text{Left quantity of milk}}{\text{Total mixture}}$$

$$= (1 - \frac{8}{64})^3 = (\frac{7}{8})^3$$

$$\frac{\text{Left quantity of milk}}{\text{Total mixture}} = \frac{343}{512}$$

$$\text{Required ratio} = 343 : (512 - 343)$$

$$= 343 : 169$$

Sol.13.(b)**Sol.14.(d)**

Ratio = Milk : Water

Initial = 2 : 1

Final = 9 : 1

Required amount of milk to be added

$$= \frac{9 - 2}{2 + 1} \times v = \frac{7}{3} v$$

Sol.15.(c)

Ratio of copper and zinc in alloy X = 7 : 3

Ratio of copper, zinc and aluminum in alloy Y = 8 : 5 : 7

Ratio - copper : zinc : Al

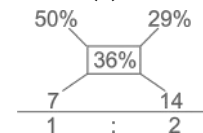
$$\text{In X} - 7 : 3 = 10 \Rightarrow 2 \times 1$$

$$\text{In Y} - 8 : 5 : 7 = 20 \Rightarrow 3$$

Ratio of copper and zinc

$$= (7 \times 2) + (8 \times 3) : (3 \times 2) + (5 \times 3)$$

$$\Rightarrow 38 : 21$$

Sol.16.(d)

Total quantity = 1 + 2 = 3 unit

Replaced quantity = 2 unit

fraction of the original mixture was

$$\text{replaced} = \frac{2}{3}$$

Sol.17.(c) According to the question

quantity of both the mixture is same

So maintaining the ratio we multiply first mixture with 11 and second with 6

Milk : Water

$$(5 : 1) \times 11$$

$$(6 : 5) \times 6$$

Milk : water

$$55 : 11$$

$$\frac{36 : 30}{91 : 41}$$

Sol.18.(a) After selling 2.5 litres of juice

from total of 19 litres, remaining quantity of juice = 19 - 2.5 = 16.5 litres

Out of 16.5 litres, the quantity of giloy

$$= 16.5 \times \frac{2}{11} = 3 \text{ litres}$$

Quantity of water = 16.5 - 3 = 13.5 litres

Let the quantity of water added be x

$$\text{ATQ, } \frac{3}{13.5 + x} = \frac{1}{5}$$

$$\Rightarrow 15 = 13.5 + x \Rightarrow x = 1.5 \text{ litres}$$

$$\text{Sol.19.(d)} \quad \text{Metal x in Alloy A} = \frac{5}{7}$$

$$\text{Metal x in alloy B} = \frac{3}{7}$$

$$\text{Metal x in alloy C} = \frac{\frac{5}{7} \times 4 + \frac{3}{7} \times 5}{4 + 5}$$



$$= \frac{\frac{20}{7} + \frac{15}{7}}{9} = \frac{35}{7 \times 9} = \frac{5}{9}$$

$$\text{Required percentage} = \frac{5}{9} \times 100 = 55\frac{5}{9}\%$$

$$\text{Sol.20.(d)} \quad 35\% = \frac{7}{20}, \quad 33\frac{1}{3}\% = \frac{1}{3}$$

Orange : Apple

$$\text{Bottle 1} \rightarrow 7 : 13 = 20 \times 3$$

$$\text{Bottle 2} \rightarrow 1 : 2 = 3 \times 20$$

Balancing the given ratio we have :

Orange : Apple

$$\text{Bottle 1} \rightarrow 21 : 39$$

$$\text{Bottle 2} \rightarrow 20 : 40$$

$$41 : 79$$

$$\text{Required percentage} = \frac{79}{41 + 79} \times 100$$

$$= \frac{79}{120} \times 100 = 65\frac{5}{6}\%$$

Sol.21.(b) since, 12 litres of the solution is taken out, then the left out solution in vessel = 32 - 12 = 20 litres

Now, Let the quantity of acid and water be 5 unit and 3 unit respectively

$$\text{ATQ, } (5 + 3) \text{ unit} = 20$$

$$8 \text{ unit} = 20$$

$$\Rightarrow 1 \text{ unit} = \frac{20}{8} = 2.5$$

$$\text{Then, } 3 \text{ unit} = 2.5 \times 3 = 7.5 \text{ litres}$$

Now, 7.5 litres of water are added to it.

Then, the required ratio = 5 : (3 + 3) = 5 : 6

Sol.22.(d) Copper : Zinc

$$\text{A} \rightarrow 3 : 4 = 7 \times 2 \times 2$$

$$\text{B} \rightarrow 5 : 9 = 14 \times 3$$

Balancing the given ratio, we have :

Copper : Zinc

$$\text{A} \rightarrow 12 : 16$$

$$\text{B} \rightarrow 15 : 27$$

$$27 : 43$$

The ratio of copper and Zinc in alloy

$$C = 27 : 43$$

Sol.23.(d) The quantity filled in container 1 = 87.5% of 40 = 35 litres

The quantity filled in container 2 = 80% of 30 = 24 litres

The quantity filled in container 3 = 90% of 20 = 18 litres

Total quantity of milk

$$= 35 \times \frac{3}{3+2} + 24 \times \frac{5}{5+1} + 18 \times \frac{7}{7+2}$$

$$= 21 + 20 + 14 = 55 \text{ litres}$$

Total quantity of water

$$= (35 + 24 + 18) - 55 = 22 \text{ litres}$$

$$\text{Required ratio} = 55 : 22 = 5 : 2$$

$$\text{Sol.24.(c)} \quad 68\% = \frac{17}{25} \text{ and } 20\% = \frac{1}{5}$$

Pulp : Water

$$\text{Fresh fruit} \rightarrow 8 : 17$$

$$\text{Dry fruit} \rightarrow 4 : 1 \times 2$$

Balancing the ratio for pulp

Pulp : Water

$$\text{Fresh fruit} \rightarrow 8 : 17$$

$$\text{Dry fruit} \rightarrow 8 : 2$$

According to the question

$$(17 + 8) = 25 \text{ unit} = 100 \text{ kg}$$

$$\text{Then, } 1 \text{ unit} = \frac{100}{25} = 4 \text{ kg}$$

$$\text{Required quantity of dry fruits} = (8 + 2)$$

$$= 10 \text{ unit} = 10 \times 4 = 40 \text{ kg}$$

Sol.25.(b)

Milk : Water

$$\text{Initial ratio} \rightarrow 5 : 3 \times 2$$

$$10 : 6$$

$$\text{New ratio} \rightarrow 2 : 3 \times 5 \quad \text{Diff.} = 9$$

$$10 : 15$$

$$\Rightarrow 16 \text{ units} = 80$$

$$\Rightarrow 9 \text{ units} = \frac{80}{16} \times 9 = 45 \text{ litres}$$

Sol.26.(a) CP of the mixture Lentil

$$= 250 \times \frac{4}{5} = ₹200$$

$$\begin{array}{cc} \text{Lentil - 1} & \text{Lentil - 2} \\ ₹90/\text{kg} & ₹220/\text{kg} \end{array}$$

$$₹200/\text{kg}$$

$$20 : 110$$

$$2 : 11$$

So, Required ratio = 2 : 11

Sol.27.(c) Initial Milk : water = 3 : 2

$$5 \rightarrow 200$$

$$1 \rightarrow 40$$

$$\text{Milk} = 120, \text{ water} = 80$$

$$\text{Final Milk : water} = 7 : 3$$

$$\text{New water} = 200 \times \frac{3}{10} = 60$$

$$\text{Loss of water} = 80 - 60 = 20 \text{ L}$$

$$\text{Removed solution} = \frac{2}{5} \rightarrow 20 = 50 \text{ L}$$

Sol.28.(c) Cost price of mixture

$$= 43.70 \times \frac{100}{115} = ₹38$$

$$\text{Price} \rightarrow ₹45 \quad ₹32$$

$$₹38$$

$$\text{Quantity} \rightarrow 6 : 7$$

$$\text{A/Q, } 7 \text{ Unit} = 24 \text{ kg}$$

$$6 \text{ unit} = \frac{24 \times 6}{7} = 20.5 \text{ kg}$$

Sol.29.(d)

$$\text{Remaining milk} = 64 \times \left(1 - \frac{8}{64}\right)^3$$

$$= 64 \times \left(\frac{7}{8}\right)^3 = 64 \times \frac{7}{8} \times \frac{7}{8} \times \frac{7}{8}$$

$$= 42.875 \approx 42.88 \text{ litres}$$

Sol.30.(b) Profit %

$$= \frac{\text{Quantity of Water}}{\text{Quantity of Pure Mango Juice}} \times 100$$

$$20 = \frac{\text{Water}}{\text{Mango Juice}} \times 100$$

$$\frac{\text{Water}}{\text{Mango Juice}} = \frac{20}{100} = \frac{1}{5} \text{ or } 1 : 5$$

Sol.31.(b)

Ratio of milk(M) to water(W) = 4 : 3

Total mixture = 70 liters $\rightarrow$ 7 unit

$$\begin{array}{ccc} \text{M} & \text{W} & \\ 4 & : & 3 \rightarrow 70 \text{ Liters} \\ 36 & : & 27 \\ +10\text{L} & & -7 \\ \hline 46 & : & 27 \end{array}$$

Hence, new ratio of milk and water = 46 : 27

Sol.32.(a) Ratio $\rightarrow$ Alcohol : Water

$$\text{Initial} \rightarrow 80 : 20$$

$$(4 : 1) \times 3$$

$$12 : 3$$

$$\text{Final} \rightarrow 85 : 15$$

$$17 : 3$$

A/Q, (12 + 3) unit = 30 litres

$$(17 - 12) \text{ unit} = \frac{30}{15} \times 5 = 10 \text{ litres}$$

Sol.33.(d)

first containers : second container

$$\text{Milk} \rightarrow 30 \quad 50$$

$$\text{water} \rightarrow 70 \quad 50$$

Total volume of both container = 200 unit

Total water in both container = 120 unit

Water percentage of final mixture

$$= \frac{120}{200} \times 100 = 60\%$$

Sol.34.(a) milk : water

$$\text{Original ratio} \rightarrow 3 : 5$$

$$\text{New ratio} \rightarrow 3 : 7$$

In both case milk same ratio

Water ratio change (7 - 5) = 2 unit

$$2 \text{ unit} = 8 \Rightarrow 1 \text{ unit} = 4$$

$$3 \text{ unit (original quantity of milk)} = 3 \times 4$$

$$= 12 \text{ litres}$$

Sol.35.(d) Let Initial quantity of mixture in the tank = 10x litres

$$\begin{array}{ccc} & 10x \text{ unit} & \\ \swarrow & & \searrow \\ \text{Initial} \rightarrow & 5x & 3x & 2x \\ \text{After replacement} \rightarrow & 5x - (10x \times \frac{5}{10}) + 5 & 3x - (10x \times \frac{3}{10}) & 2x - (10x \times \frac{2}{10}) + 3 \\ & 5x & 3x - 3 & 2x + 1 \end{array}$$

$$\text{A/q, } 5x - (3x - 3) = 25$$

$$\Rightarrow 2x = 22 \Rightarrow x = 11$$

Hence, initial quantity of mixture in the tank (in litres) = 10 $\times$ 11 = 110

Sol.36.(b) Let the ratio be 4x, 6x, 10x

A/Q,

$$(4x + 6) : (6x + 12) : (10x + y) = 6 : 10 : 16$$



For P and Q

$$\frac{4x+6}{6x+12} = \frac{3}{5} \Rightarrow 20x + 30 = 18x + 36$$

$$\Rightarrow 2x = 6 \Rightarrow x = 3$$

For Q and R

$$\frac{6x+12}{10x+y} = \frac{5}{8} \Rightarrow \frac{30}{30+y} = \frac{5}{8}$$

$$\Rightarrow 240 = 150 + 5y$$

$$\Rightarrow 5y = 90 \Rightarrow y = 18$$

Sol.37.(c)

$$\text{Final quantity of syrup} = 200\left(1 - \frac{1}{10}\right)^2$$

$$= 200 \times \frac{81}{100} = 162\text{L}$$

$$\text{Quantity of water} = 200 - 162 = 38\text{L}$$

$$\text{Ratio of Water to Syrup} = 38\text{L} : 162\text{L} = 19 : 81$$

Sol.38.(a) Let total mixture = $16x$

According to question,

$$\text{Apple juice} = \left(\frac{16x-24}{16} \times 7\right)$$

$$\text{Mango juice} = \left(\frac{16x-24}{16} \times 5 + 10\right)$$

A/Q,

$$\Rightarrow \left(\frac{16x-24}{16} \times 7\right) - \left(\frac{16x-24}{16} \times 5 + 10\right)$$

$$= 14$$

$$\Rightarrow \frac{16x-24}{16} \times 2 = 14 + 10$$

$$\Rightarrow \frac{16x-24}{16} = 12 \Rightarrow x = 12 + \frac{24}{16}$$

$$16x = 12 \times 16 + 24 \Rightarrow 216$$

Hence, total mixture 216 liters

Sol.39.(a)

$$\text{Remaining water} = 7\frac{3}{4} - 2 \text{ (3.375)}$$

$$= 7.75 - 6.75 = 1 \text{ liter}$$

Sol.40.(b)Container $\rightarrow$ A B

$$\text{Ratio} \rightarrow \text{Alcohol : water} \quad \text{Alcohol : water}$$

$$2 : 1 \quad 4 : 3$$

Now 9 liters of mixture from container A

$$\text{Alcohol} = \frac{2}{3} \times 9 = 6$$

$$\text{Water} = \frac{1}{3} \times 9 = 3$$

14 liters of mixture from container B

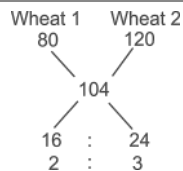
$$\text{Alcohol} = \frac{4}{7} \times 14 = 8$$

$$\text{Water} = \frac{3}{7} \times 14 = 6$$

Ratio of alcohol to water in the new mixture $\rightarrow 6 + 8 : 3 + 6 = 14 : 9$ **Sol.41.(c)**

$$\text{Cost price of mixture} = \frac{130 \times 100}{125}$$

$$= ₹104 \text{ per kg.}$$



Ratio = 2 : 3

Sol.42.(c)

$$\text{Initial milk} = 120 \times \frac{5}{8} = 75 \text{ litres.}$$

$$\text{Initial water} = 120 \times \frac{3}{8} = 45 \text{ litres.}$$

According to the question,

$$\frac{75+x}{45+x} = \frac{3}{2} \Rightarrow x = 15 \text{ litres}$$

Sol.43.(a) Ratio $\rightarrow$ Alcohol : Water

$$1\text{st} \rightarrow 5 : 3$$

$$2\text{nd} \rightarrow 5 : 9$$

A/Q, 6 unit = 24 litre

$$1 \text{ unit} = 4 \text{ litre}$$

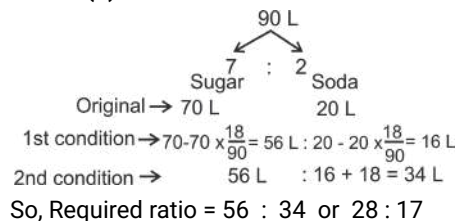
$$8 \text{ unit} = 32 \text{ litre}$$

Initial quantity of mixture = 32 litre

Sol.44.(a)Ratio $\rightarrow$ Juice : Water

$$2 : 3$$

$$3 : 2$$

On mixing $\rightarrow 5 : 5$ or $1 : 1$ **Sol.45.(b)****Sol.46.(c)** Total mixture = 60 ltr

Juice : Water = 40 : 20

$$\Rightarrow \frac{\text{Juice}}{\text{Water}} = \frac{2}{3} \Rightarrow \frac{40}{20+x} = \frac{2}{3}$$

$$\Rightarrow 120 = 40 + 2x = 2x = 80 = x = 40$$

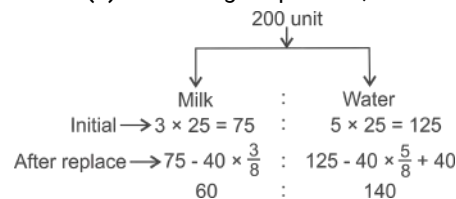
Sol.47.(b) Total mixture = $6x + 5x = 11x$

According to question

$$\frac{6x-10}{5x+9} = \frac{2}{3} \Rightarrow 18x - 30 = 10x + 18$$

$$\Rightarrow 18x - 10x = 30 + 18 \Rightarrow 8x = 48 \Rightarrow x = 6$$

$$\text{Total mixture} = 11 \times 6 = 66 \text{ L}$$

Sol.48.(d) According to question,

$$\text{Percentage of acid} = \left(\frac{60}{200}\right) \times 100 = 30\%$$

Sol.49.(b)Ratio $\rightarrow$ Milk : water

$$\text{Initial} \rightarrow 3 : 5$$

$$\text{Final} \rightarrow 3 : 6$$

Now according to question ,

$$(3 + 5) \text{ unit} = 32 \text{ liter}$$

$$(6 - 5) \text{ unit} = \frac{32}{8} \times 1 = 4\text{L}$$

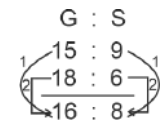
Sol.50.(c) Gold : Silver sum

$$5 : 3 \rightarrow 8 \times 3$$

$$3 : 1 \rightarrow 4 \times 6$$

$$4 : 2 \rightarrow 6 \times 4$$

After equalizing the ratio of the three alloys -



Therefore, the required ratio = 2 : 1

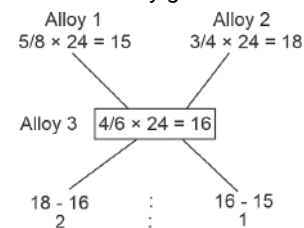
According to the question,

$$1 \text{ unit} \rightarrow 40 \text{ kg}$$

$$2 \text{ unit} \rightarrow \frac{40 \times 2}{1} = 80 \text{ kg}$$

Another method :-

Consider only gold in the mixtures -

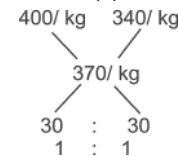


Therefore, the required ratio = 2 : 1

According to the question,

$$1 \text{ unit} \rightarrow 40 \text{ kg}$$

$$2 \text{ unit} \rightarrow \frac{40 \times 2}{1} = 80 \text{ kg}$$

Sol.51.(a)**Sol.52.(a) Given :** Solution = 150 litres

$$\text{Case I :- Quantity of salt} = 150 \times \frac{40}{100}$$

$$= 60 \text{ L}$$

$$\text{Quantity of water} = 150 - 60 = 90 \text{ L}$$

According to question,

$$\Rightarrow \frac{60}{90+x} = \frac{20}{80}$$

$$\Rightarrow 240 = 90 + x \Rightarrow x = 150$$

$$\text{Case II :- } \frac{60+x}{90+150} = \frac{1}{1}$$

$$\Rightarrow x = 240 - 60 = 180 \text{ L}$$

Sol.53.(b) Let number = $8x, 9x$

If 8 is added to each the new ratio becomes 17 : 19

$$\Rightarrow \frac{8x+8}{9x+8} = \frac{17}{19}$$

$$\Rightarrow 152x + 152 = 153x + 136 \Rightarrow x = 16$$

$$1\text{st no} = 8 \times 16 = 128$$

$$2\text{nd no.} = 9 \times 16 = 144$$



Sol.54.(b)

$$\begin{array}{c} 80L \\ \swarrow \quad \searrow \\ 3M : 1W \\ \text{Original} \rightarrow 60L : 20L \\ \text{1st condition} \rightarrow 60 - 60 \times \frac{12}{80} = 51L : 20 - 20 \times \frac{12}{80} = 29L \\ \text{2nd condition} \rightarrow 51 - 51 \times \frac{12}{80} = 35.35L : 29 - 29 \times \frac{12}{80} = 24.65L \end{array}$$

Now, Required % = $\frac{24.65}{80} \times 100 = 30.81\%$

Sol.55.(b)

Ratio $\rightarrow$ Milk : Water
 Initial $\rightarrow 3 : 1$
 Final $\rightarrow 3 : 5$
 A/Q, (3 + 1) unit = 40 L
 (5 - 1) unit = $\frac{40}{4} \times 4 = 40$ L

Sol.56.(d) Final Quantity

= Initial Quantity $\times (1 - \frac{\text{Amount Removed}}{\text{Total Volume}})^n$
 = $100 \times (1 - \frac{10}{100})^3$
 = $100 \times (\frac{90}{100})^3 = \frac{729}{10} = 72\frac{9}{10}$

Sol.57.(d) $10\% = \frac{1}{10}$

Initial Volume of Acid = $\frac{100}{5} \times 3 = 60$ L

Final acid = $60 \times \frac{9}{10} \times \frac{9}{10} = 48.6$ L

Required Percentage = $\frac{48.6}{100} \times 100$
 = 48.6%

Sol.58.(a) Let original no of red, blue and green marbles = 4x, 5x, 6x

A/Q, 4x - 6, 5x, 6x + 3,
 then new ratio would be 2 : 5 : 7

$\Rightarrow \frac{4x - 6}{5x} = \frac{2}{5} \Rightarrow 5(4x - 6) = 2(5x)$

$\Rightarrow 20x - 30 = 10x \Rightarrow 10x = 30 \Rightarrow x = 3$

Initial Red marbles = $4 \times 3 = 12$

Sol.59.(a)

$$\begin{array}{c} 150L \\ \swarrow \quad \searrow \\ 3A : 7W \\ \text{Original} \rightarrow 45L : 105L \\ \text{After removing} \rightarrow 45 \times \frac{130}{150} = 39L : 105 \times \frac{130}{150} = 91L \\ \text{After add} \rightarrow 39 + 20 \times \frac{40}{100} = 47L : 91 + 20 \times \frac{60}{100} = 103L \end{array}$$

Required percentage = $\frac{47}{150} \times 100$
 = 31.33%

Sol.60.(a) According to the question:

Final amount of acid

$= 64 - (\frac{10}{80} \times 64) + (\frac{1}{5} \times 10) = 64 - 8 + 2$
 = 58 L

Water = $16 - (\frac{10}{80} \times 16) + (\frac{4}{5} \times 10)$
 = $16 - 2 + 8 = 22$ L

Final ratio = $\frac{\text{Acid}}{\text{Water}} = \frac{58}{22} = \frac{29}{11}$
 = 29 : 11

Sol.61.(d) Final Quantity =

Initial Quantity $\times (1 - \frac{\text{Amount Removed}}{\text{Total Volume}})^n$
 = $136 \times (1 - \frac{17}{136})^2 = 136 \times (\frac{7}{8})^2$
 = $17 \times \frac{49}{8} = \frac{833}{8} = 104\frac{1}{8}$ litres

Sol.62.(c) Remaining spirit

$= 54 \times (1 - \frac{9}{54})^3 = 54 \times (\frac{5}{6})^3$
 $= 54 \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} = 31.25$ L

Sol.63.(b)

Final quantity of milk = $100(1 - \frac{20}{100})^2$

$= 100 \times \frac{16}{25} = 64$ L

Quantity of water = $100 - 64 = 36$ L

Ratio of Water to milk = 36L : 64L = 9 : 16

Sol.64.(a)

Given:- Selling Price of mixture = ₹121 per kg

Cost Price of mixture = $\frac{121}{110} \times 100$

= ₹110 per kg

Using Alligation Method:-

$$\begin{array}{c} 75 \quad \quad 125 \\ \swarrow \quad \searrow \\ 110 \\ \swarrow \quad \searrow \\ 15 \quad \quad 35 \end{array}$$

Ratio = 15 : 35

= 3 : 7

Sol.65.(b) CP = $84 \times \frac{100}{120} = \text{Rs.}70$

$$\begin{array}{c} \text{1st type} \quad \quad \text{2nd type} \\ \text{₹60/kg} \quad \quad \text{₹90/kg} \\ \swarrow \quad \quad \searrow \\ \text{₹70/kg} \\ \swarrow \quad \quad \searrow \\ 20 \quad \quad 10 \\ 2 \quad \quad 1 \end{array}$$

So, Ratio = 2 : 1

Sol.66.(a)

CP of coffee = $138 \times \frac{100}{115} = \text{₹}120/\text{kg}$

$$\begin{array}{c} \text{₹}100/\text{kg} \quad \quad \text{₹}140/\text{kg} \\ \swarrow \quad \quad \searrow \\ \text{₹}120/\text{kg} \\ \swarrow \quad \quad \searrow \\ 20 \quad \quad 20 \\ (1) \quad \quad (1) \end{array}$$

$\Rightarrow 1$ units = 80 kg

So, the dearer variety of coffee added
 = 80 kg

Sol.67.(c)

Final quantity of milk = $120(1 - \frac{30}{120})^2$

$= 120 \times \frac{9}{16} = 67.5$ L

Quantity of water = $120 - 67.5 = 52.5$ L

Ratio of Water to milk = 52.5 L : 67.5 L = 7 : 9

Sol.68.(a)

$$\begin{array}{c} \text{Water} : \text{Acid} \\ 7 : 3 \\ \times 1 \quad \times 1 \\ \hline 7 : 3 \\ \times 7 \quad \times 7 \\ \hline 49 : 21 \end{array}$$

10 unit = 150L
4 unit = $4 \times 15 = 60$ L

Acid to be added = 60L

Sol.69.(b) $14\frac{1}{7}\% = \frac{1}{7}$

Cost price of the mixture = $80 \times \frac{7}{8} = \text{₹}70/\text{kg}$

$$\begin{array}{c} \text{Price} \rightarrow \text{₹}65 \quad \quad \text{₹}95 \\ \swarrow \quad \quad \searrow \\ \text{₹}70 \\ \swarrow \quad \quad \searrow \\ \text{Quantity} \rightarrow 25 : 5 \\ 5 : 1 \end{array}$$

5 unit = 45kg

1 unit = 9kg

Sol.70.(c)

$$\begin{array}{c} \text{Ratio} \rightarrow \text{Juice} : \text{Water} \\ 4 : 3 \\ \times 5 \quad \times 5 \\ \hline 20 : 15 \\ 5 : 4 \\ \times 4 \quad \times 4 \\ \hline 20 : 16 \end{array}$$

1 unit = 21 litres

Initial quantity of mixture = 7×21

= 147 litres

Sol.71.(b) Acid : Water
 (7 : 3) $\times 4$
 28 : 12
 (4 : 5) $\times 7$
 28 : 35

Now according to question,

$\Rightarrow (28 + 12)$ unit $\rightarrow 60$

$\Rightarrow 1$ unit = $\frac{60}{40} = 1.5$ L

$\Rightarrow (35 - 12)$ unit $\rightarrow 23 \times 1.5 = 34.5$ liters

Sol.72.(b) Cost of the mixture per kg

$= \frac{160 \times 2 + 210 \times 3}{2 + 3} = \frac{950}{5} = \text{₹}190$

Sol.73.(b) For can 1 :- acid : water = 5 : 3

Acid = $\frac{5}{8}x$, water = $\frac{3}{8}x$

For can 2 :- acid : water = 2 : 3

Acid = $\frac{2}{5}y$, water = $\frac{3}{5}y$

Total acid = $\frac{5}{8}x + \frac{2}{5}y$

Total water = $\frac{3}{8}x + \frac{3}{5}y$

According to the question,

$\frac{5}{8}x + \frac{2}{5}y = \frac{3}{8}x + \frac{3}{5}y$



$$\frac{2}{8}x = \frac{1}{5}y$$

$$10x = 8y \Rightarrow x : y = 4 : 5$$

Required ratio = 4 : 5

Sol.74.(a) According to question,

$$\Rightarrow 10 \text{ units} = 40 \text{ litres}$$

$$\Rightarrow 1 \text{ unit} = 4 \text{ litres}$$

Alcohol = 12 litres and water = 28 litres

Now,

Required ratio $\rightarrow 12 : (28 + 40)$

$$= 12 : 68 = 3 : 17$$

Sol.75.(d)

Grape juice : Pineapple juice

$$\begin{array}{l} \text{Initial} - 13 : 11 \\ \text{Final} - 13 : 14 \end{array} \Rightarrow 3 \text{ unit} = 8 \text{ ltr}$$

As per figure,

$$1 \text{ unit} = \frac{8}{3} \text{ ltr.}$$

Amount of pineapple juice in the remaining mixture before adding new pineapple juice

$$= 11 \text{ unit} = 11 \times \frac{8}{3} = \frac{88}{3} \text{ ltr}$$

Now, pineapple juice in 8 ltr mixture

$$= \frac{11}{11 + 13} \times 8 = \frac{11}{3} \text{ ltr}$$

So, total pineapple juice initially in the

$$\text{can} = \frac{88}{3} + \frac{11}{3} = \frac{99}{3} = 33 \text{ ltr.}$$

Sol.76.(b) Let the three varieties of rice

be $X = 4a$, $Y = a$ and $Z = 3a$

According to question,

$$\Rightarrow 8a = 4 \text{ kg} \Rightarrow a = 0.5 \text{ kg}$$

Then,

$$X = 4 \times 0.5 = 2 \text{ kg}, Y = 0.5 \text{ kg} \text{ and } Z = 3 \times 0.5 = 1.5 \text{ kg}$$

Now,

$$\text{Cost of the packet} = 2 \times ₹ 2 + 0.5 \times ₹ 4 + 1.5 \times ₹ 6 = 4 + 2 + 9 = ₹ 15$$

Sol.77.(a)

Price $\rightarrow 84 : 96 : x$

Quantity $\rightarrow 1 : 1 : 2$

Total $\rightarrow 84 : 96 : 2x$

According to question,

$$84 + 96 + 2x = (1 + 1 + 2) \times 99$$

$$2x = 396 - 180$$

$$x = \frac{216}{2} = ₹ 108$$

Sol.78.(a) Given: concentration in new solution = 30%

Acid $\rightarrow A : B : C$

Ratio $\rightarrow 3 : 5 : a$

Concentration(%) $\rightarrow 20 : 30 : 40$

Deviation $\rightarrow -10 : 0 : +10$

According to question

$$\frac{20 \times 3 + 30 \times 5 + 40 \times a}{3 + 5 + a} = 30$$

$$\frac{60 + 150 + 40a}{8 + a} = 30$$

$$210 + 40a = 240 + 30a$$

$$10a = 30 \Rightarrow a = 3$$

Sol.79.(c) By using alligation method

Let cost price of mixture = ₹x

$$\begin{array}{r} \text{₹4} \quad \quad \quad \text{₹6} \\ \quad \quad \quad \times \\ \quad \quad \quad 5\text{kg} \quad \quad 3\text{kg} \\ x = \frac{4 \times 5 + 6 \times 3}{5 + 3} = \frac{38}{8} = ₹4.75 \end{array}$$

Sol.80.(b) Cost of type 1 rice (1 g)

$$= \frac{250}{5} = ₹50$$

$$\text{Cost of type 2 rice (1 g)} = \frac{280}{5} = ₹56$$

According to the question,

Net cost of resultant mixture

$$= \frac{50 \times 7 + 56 \times 8}{7 + 8} = \frac{350 + 448}{15} = \frac{798}{15} = ₹53.20$$

Sol.81.(c) Let the third variety of tea powder be x

According to the question,

$$250 \times 1 + 300 \times 1 + 2 \times x = 350 \times 4$$

$$550 + 2x = 1400$$

$$2x = 850 \Rightarrow x = ₹ 425 / \text{kg}$$

Sol.82.(d)

Let second type wheat ₹x per kg

$$15 \times 10 + x \times 5 = 30 \times 20$$

$$150 + 5x = 600 \Rightarrow 5x = 450 \Rightarrow x = ₹90$$

Sol.83.(d) Let the quantity of water and mango a in juice be $9a$ and $7a$ respectively,

According to the question,

$$16a = 160 \Rightarrow a = 10$$

$$9a = 90 \text{ and } 7a = 70$$

$$\text{Then, } \Rightarrow \frac{90 + x}{70 + 3x} = \frac{13}{14}$$

$$\Rightarrow 1260 + 14x = 910 + 39x$$

$$\Rightarrow 350 = 25x \Rightarrow x = \frac{350}{25} = 14$$

Quantity of the new mango juice

$$= (90 + 14) + (70 + 3 \times 14) = 216 \text{ litre}$$

Sol.84.(d) Selling Price of sugar = ₹75

$$\text{Cost price of mixed sugar} = 75 \times \frac{100}{120} = ₹62.5$$

Now By alligation Method,

$$\begin{array}{r} \text{CP} \rightarrow \text{₹78} \quad \quad \quad \text{₹46} \\ \quad \quad \quad \times \\ \quad \quad \quad \text{₹62.5} \\ \text{Quantity} \rightarrow \text{₹16.5} \quad : \quad \text{₹15.5} \\ \quad \quad \quad 33 \quad \quad : \quad 31 \end{array}$$

Sol.85.(c) By using allegation method

$$\begin{array}{r} \text{Price} \rightarrow \text{₹26} \quad \quad \quad \text{₹x} \\ \quad \quad \quad \times \\ \quad \quad \quad \text{₹35} \\ \text{Quantity} \rightarrow 10 \quad : \quad 6 \end{array}$$

$$\Rightarrow \frac{26 \times 10 + 6x}{16} = 35$$

$$\Rightarrow 260 + 6x = 35 \times 16$$

$$\Rightarrow 6x = 300 \Rightarrow x = 50$$

$\therefore$ price of good quality wheat per kg = ₹50

Sol.86.(b)

Ratio $\rightarrow$ Alcohol : Water

Initial $\rightarrow 3 : 17 \rightarrow$ total = 20 unit

Final $\rightarrow 1 : 1) \times 17 = 17 : 17$

Part of the alcohol that was added

$$= 17 - 3 = 14 \text{ units}$$

$$20 \text{ unit} = 500 \text{ ml}$$

$$14 \text{ unit} = \frac{500}{20} \times 14 = 25 \times 14 = 350 \text{ ml}$$

Sol.87.(a) Let x liter water be added

According to the question,

$$\frac{60}{60 + x} = \frac{\frac{3}{2}}{\frac{15}{8}}$$

$$\frac{60}{60 + x} = \frac{3}{2} \times \frac{8}{15} \Rightarrow \frac{60}{60 + x} = \frac{4}{5}$$

$$300 = 240 + 4x \Rightarrow 4x = 60 \Rightarrow x = 15$$

Sol.88.(c) Required temperature

$$= \frac{(2 \times 100) + (18 \times 32)}{2 + 18}$$

$$= \frac{200 + 576}{20} = 38.8^\circ \text{C}$$

